

01 - Requirements: HCNR201 Selectable Output Analog Isolator - Rev B2

Revision summary

Rev B2 adopts a simplified host-power protection chain consisting only of an AO3401A P-channel MOSFET for reverse-polarity protection and an FB1 ferrite bead for conducted-noise filtering. The previous PPTC and input TVS are removed from the production design.

Parameter	Rev B2 requirement
Sensor analog input	0-5.0 V DC from a conditioned sensor module
Output modes	0-2.4 V, 0-3.0 V, or 0-4.5 V selected by a mutually exclusive SP3T switch
Host input supply	Regulated 5 V source, preferably with upstream current limiting
Reverse-polarity protection	Q1 AO3401A; drain to 5V_IN, source to protected rail, gate pull-down 100 k Ω
Input surge protection	No on-board TVS in Rev B2
Overcurrent protection	No on-board PPTC in Rev B2
Conducted-noise filtering	FB1 ferrite bead between Q1 and PS1; 0 Ω validation option may share the footprint
Isolated converter	YLPTEC IF0505S-1WR3, regulated isolated 5 V
Analog amplifiers	OPA317 preferred for U1 and U2
Isolation rule	GND_HOST and GND_ISO must remain isolated
Calibration	Fixed precision resistors and two-point software calibration recommended

Host-supply responsibility

Because Rev B2 omits the PPTC, the upstream USB port, regulator, or system power supply should include appropriate current limiting. Because the input TVS is omitted, the module is intended primarily for short, internal 5 V wiring or installations where surge and ESD protection are provided elsewhere.

Output-mode requirements

Mode	Host class	Nominal feedback	Nominal output FS
MODE 24	ESP32-safe / sensitive 3.3 V ADC	80.6 k Ω	2.4 V
MODE 33	Generic 3.3 V ADC	101 k Ω	3.0 V
MODE 50	5 V ADC / PLC	150 k Ω + 1.8 k Ω	4.5 V